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Study on the Pseudo Threshold Pressure Gradient Model of Heavy Oil Reservoirs Considering CO₂ Dissolution Viscosity Reduction and Thermal Effects

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Abstract

This study investigates Xinjiang oilfield heavy oil as the research subject, employing a high-temperature, high-pressure rheometer to analyze changes in crude oil viscosity under varying temperatures after CO₂ dissolution, thereby exploring the synergistic effects of CO₂ and thermal action on heavy oil viscosity reduction. Additionally, core flooding experiments were conducted to examine the influence of this synergy on the flow behavior of heavy oil in reservoirs. The impact of CO₂ dissolution at different temperatures on the pseudo threshold pressure gradient (PTPG) of heavy oil was tested, and the relationship between mobility, PTPG, and activation temperature was established. The results indicate that the viscosity of heavy oil decreases with increasing CO₂ solubility. However, the viscosity-reducing effect of CO₂ diminishes at higher temperatures. Through rheological experiments, constitutive Eq.s for heavy oil under different CO₂ solubilities were developed. Core flow experiments yielded expressions for the pseudo-TPG and activation temperature of heavy oil under combined CO₂ and thermal effects. A small amount of CO₂ can achieve flow performance equivalent to that at elevated temperatures, significantly reducing the thermal energy demand for heavy oil recovery. Furthermore, these expressions enable more accurate predictions of the maximum drainage radius and optimal thermal stimulation transition temperature in heavy oil reservoirs. This study establishes models for the TPG and activation temperature of heavy oil under the synergistic action of CO₂ dissolution and thermal effects, clarifying the influence of CO₂ and heat on the PTPG. These models improve the prediction of the maximum drainage radius and thermal transition temperature in heavy oil reservoirs. The findings provide theoretical guidance for CCUS and CO₂-assisted steam flooding EOR technologies in similar Xinjiang oilfield blocks.

Keywords: Pseudo threshold pressure gradient (PTPG), CCUS, EOR, CO₂-Assisted;heavy oil

Introduction

China possesses abundant heavy oil resources (Wang et al. 2016; Frances J. Hein 2017; Zhang et al. 2023), whose high-viscosity characteristics lead to significant non-Darcy flow behavior during reservoir

seepage, requiring overcoming a threshold pressure gradient to achieve effective flow (Dong et al. 2013). Currently, thermal viscosity reduction technologies are widely employed in the industry to enhance heavy oil mobility (Sayed Gomaa et al. 2024; Alvarado V. 2010; Nasr T.N., and O.R. Ayodele. 2005). Among these, CO₂-assisted thermal recovery has emerged as a key method for improving production efficiency during the late stages of steam stimulation. By leveraging the synergistic effects of thermal action and CO₂ dissolution-induced viscosity reduction, this approach significantly enhances heavy oil flow efficiency in porous media (Dong et al. 2019). However, the nonlinear seepage mechanisms of heavy oil under thermo-chemical coupling conditions and the dynamic evolution of pseudo-threshold pressure gradients remain unclear, which constrains the optimal design of CCUS technology and CO₂ displacement processes.

The high viscosity of heavy oil fundamentally originates from the three-dimensional network structure formed by its heavy components (resins and asphaltenes) through hydrogen bonding and molecular chain entanglement (Cui et al. 2017). CO₂ dissolution effectively disrupts this network structure through a dilution effect that reduces intermolecular interactions between asphaltenes, thereby breaking down the spatial network. Additionally, CO₂ reduces the surface tension of heavy oil and weakens its adsorption on rock pore surfaces, consequently enhancing the mobility of heavy oil in reservoirs (Cao et al. 2021; Su and Yue 2020). The non-Newtonian fluid characteristics of heavy oil and its threshold pressure gradient effect constitute critical research challenges for enhanced oil recovery. Yang Jian (Jian et al. 2018) demonstrated that heavy oil exhibits Bingham fluid behavior under low-temperature conditions, requiring overcoming the threshold pressure gradient to initiate flow. During steam flooding processes, formation temperature fields can be categorized into latent heat zones, sensible heat zones, and unheated zones. The temperature-threshold pressure gradient coupling model developed by Yang reveals the significant impact of thermal effects on seepage patterns. Xin Xiankang (Xin et al. 2017) further verified through experiments and simulations that heavy oil viscosity shows negative correlations with both temperature and asphaltene content, while the threshold pressure gradient decreases with either reduced viscosity or increased permeability. This relationship follows a power-law dependence on flow velocity and is significantly influenced by viscosity and permeability. Liu Wenchao (Liu 2020) emphasized that traditional Darcy's law has limitations in describing flow behavior under threshold pressure gradient conditions, necessitating the introduction of boundary condition constraints to accurately characterize seepage in heterogeneous formations. Fan Tinggen's (Fan et al. 2022) productivity model confirmed that bottomhole pressure represents the dominant factor in heavy oil development, with the threshold pressure gradient's effect on increasing flow resistance being secondary. Ke Wenli's (Ke et al. 2022) numerical simulations further indicated that threshold pressure gradients exacerbate flow resistance during water flooding processes, leading to uneven remaining oil distribution and elevated saturation levels. Hu Ruien (Hu 2015) systematically studied the rheological properties of CO₂-dissolved crude oil, demonstrating that the viscosity-temperature relationship of the mixed fluid still follows an exponential decline pattern, and that CO₂ dissolution does not alter the non-Newtonian fluid characteristics of heavy oil.

Currently, research on the flow behavior of CO₂-dissolved heavy oil in porous media remains relatively limited. To evaluate the seepage characteristics of different crude oils with dissolved CO₂ under reservoir conditions, this study systematically investigates the influence of CO₂ dissolution on the viscoelastic and shear-thinning properties of heavy oil through viscosity-temperature measurements, rheological testing, and unidirectional core flooding experiments. The work elucidates the threshold pressure and pseudo-threshold pressure gradient evolution mechanisms governing nonlinear seepage in CO₂-heavy oil systems within porous media. Furthermore, we establish a predictive model for the initiation pressure in CO₂-assisted thermal recovery based on coupled rheological and seepage data. These findings provide crucial theoretical support for optimizing CO₂ injection parameters in CCUS applications and enhancing steam displacement efficiency in analogous Xinjiang oilfield reservoirs.

Experimental procedures

Materials

The heavy oil samples used in this study were obtained from Xinjiang Oilfield. After collection, the samples underwent degassing and dehydration treatment. The four components (SARA analysis: saturates, aromatics, resins, and asphaltenes) were determined according to the NB/SH/T 0590-2010 standard. The basic parameters of the crude oil are presented in Table 1. The formation water used in the experiments is shown in Table 2. High-pressure compressed CO₂ with a purity of 99.9% was employed as the experimental gas. Artificial core samples with dimensions of 10 cm in length and 2.5 cm in diameter were used, exhibiting an average porosity of 25.5% and an average permeability of approximately $2000 \times 10^{-3} \mu\text{m}^2$.

Table 1—SARA components and basic properties of heavy and extra-heavy crude oils

Oil type	Density	Viscosity	SARA components(wt.%)			
	(g·mL ⁻¹ , 20°C)	(mPa·s, 25°C)	Saturate	Aromatic	Resin	Asphaltene
Heavy oil	0.918	9729.85	59.63	27.16	12.07	1.14

Table 2—Mineralization level of experiment water

Cation (mg/L)				Anion (mg/L)			Total Dissolved
K ⁺ +Na ⁺	Mg ²⁺	Ca ²⁺	Cl ⁻	SO ₄ ²⁺	HCO ₃ ⁻	CO ₃ ²⁻	Solids (mg/L)
3425.32	43.52	106.91	1783.34	237.47	1791.80	101.63	7489.99

The instruments used in the experiment included a constant-temperature oven (maximum working temperature: 300°C), a core holder (maximum pressure resistance: 70 MPa), an ISCO displacement pump, a piston, a six-way valve, a pressure acquisition system, a high-temperature and high-pressure rheometer, various pipelines, and glassware.

Experiment Method

Viscosity-Temperature and Rheological Property Experiments. We investigated the effects of CO₂ injection on the rheological behavior of crude oil using a high-temperature, high-pressure rheometer system. During the experiment, crude oil was first mixed with a solvent at preset ratios (20, 30, 40 m³/m³) and homogenized using a mechanical stirrer at 60 r/min for 60 minutes. The homogeneous samples were then transferred to the rheometer test chamber, where viscosity-temperature curve and rheological curve measurements were sequentially conducted under sealed pressurization. The viscosity-temperature curve test employed a continuous heating mode (25–130°C, heating rate 0.5°C/min) with a constant shear rate of 7.34 s⁻¹. The rheological curve test was performed under isothermal conditions (25°C) via dynamic shear scanning, covering a shear rate range of 0–180 s⁻¹. To ensure data reliability, the experiment was repeated by adjusting CO₂ dissolution levels (0% to saturated solubility), with each condition tested in triplicate to minimize operational errors. The system ultimately recorded the rheological response characteristics of crude oil under varying CO₂ concentrations.

Pseudo threshold pressure test. The experiment simulated reservoir stress conditions (confining pressure: 7 MPa, backpressure: 5 MPa) using a high-temperature and high-pressure core flooding system to reveal the nonlinear seepage mechanisms of CO₂-heavy oil systems through multi-parameter coupling control, with the experimental schematic shown in Fig. 1. The specific experimental procedure was as follows: ① Core oil saturation: The core was placed in a core holder and saturated with oil at a flow rate of 0.1 ml/min according to the process shown in Fig. 1. When the oil production at the outlet equaled the injection volume, flooding was continued for 2 PV to calculate the oil saturation. ② The valves at both ends of the core holder were closed, and the crude oil was aged at 40°C for 24 hours before commencing the flooding experiment. ③ Different flooding rates (0.1–0.5 ml/min) were set according to the experimental protocol, and the equilibrium pressures during crude oil flow at different rates were recorded. ④ The experimental temperature was varied, and the equilibrium pressures during crude oil flow at different rates were recorded. ⑤ The relationship curve between seepage velocity and pressure gradient was plotted, and a regression Eq. was fitted. The intercept of the regression curve with the coordinate axis was calculated as the pseudo-threshold pressure gradient.

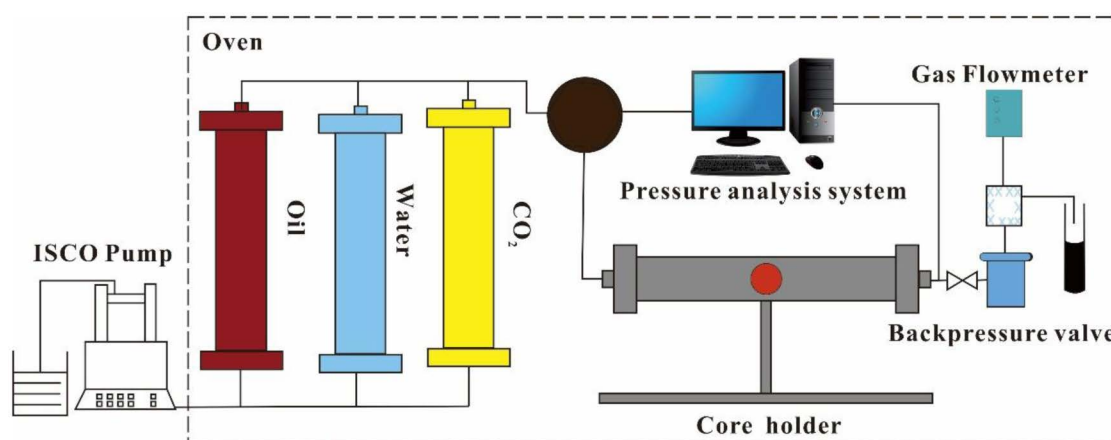


Figure 1—Core percolation experiment flowchart

Results and discussion

The Effect of CO₂ on the Viscosity-Temperature Characteristics of Crude Oil

Temperature exerts a profound influence on the viscosity of heavy oil, as evidenced by the nonlinear temperature-viscosity relationships in Fig. 2(a), where viscosity decreases markedly with rising temperature. This behavior stems from thermal disruption of π - π bonds and hydrogen bonds within asphaltene-resin aggregates (Arinina et al. 2019; Jin et al. 2022), transforming dense structures into mobile sol-like configurations. Below the anomalous temperature (51.8°C), the oil exhibits Bingham fluid characteristics, transitioning to Newtonian behavior above this threshold (Anastasiya, and Sergey 2022). Fig. 2(b) reveals a bilinear log(viscosity)-temperature relationship with a distinct inflection point: below this point, viscosity drops rapidly (linear in log scale), while above it, the decline moderates as Newtonian flow dominates. CO₂ injection preserves this bilinear trend but systematically lowers the inflection temperature—reducing it to 49.8°C, 47.8°C, and 45.7°C at CO₂ concentrations of 20, 30, and 40 m³/m³, respectively. This shift occurs because dissolved CO₂ weakens intermolecular forces between asphaltenes, lowering both the anomalous temperature and the energy barrier for flow, thereby enhancing mobility at operational temperatures.

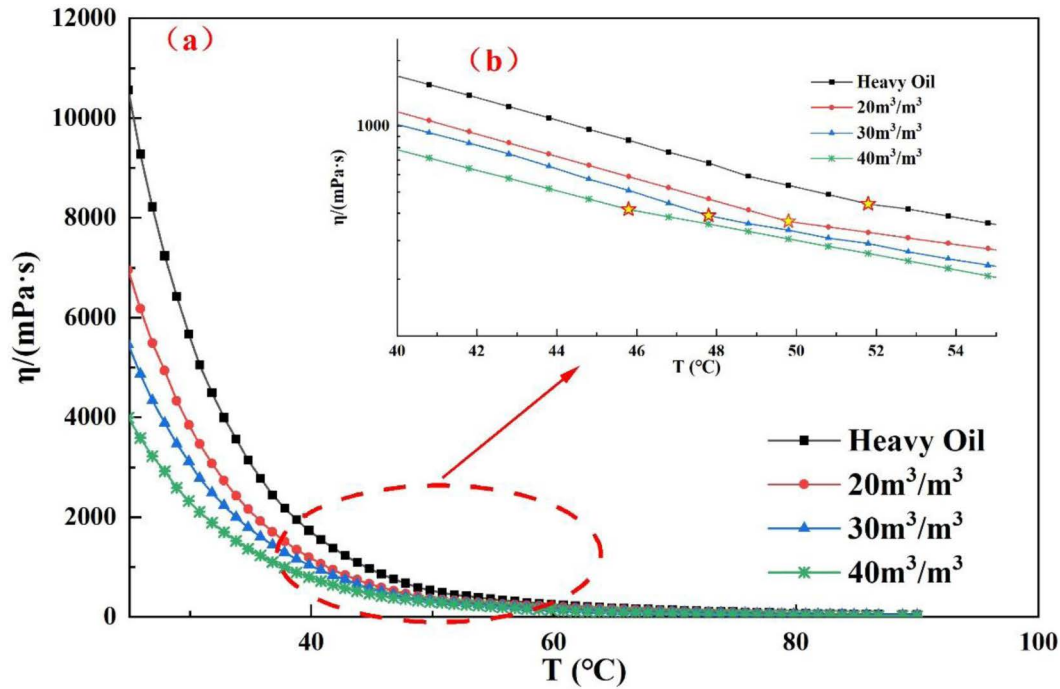


Figure 2—Viscosity-Temperature Curves and Anomalous Temperature Points at Different Solubility Levels

To quantitatively characterize the temperature-dependent viscosity reduction of heavy oil, we introduce the viscosity reduction ratio θ , defined as the relative difference between the viscosity μ_1 at temperature T_1 and the baseline viscosity μ_0 at 25°C, as expressed in Eq. (1):

$$\theta = \frac{u_0 - \mu_1}{\mu_0} \quad (1)$$

Where, θ is viscosity reduction ratio, %; u_0 baseline viscosity at reference temperature T_0 , mPa·s; u_1 is baseline viscosity at reference temperature T_1 , mPa·s.

Fig. 3 presents the viscosity reduction ratio curves for different CO₂ injection volumes. The results demonstrate that the viscosity reduction ratio (θ) of heavy oil increases with rising temperature, while at any given temperature, θ values are consistently higher for CO₂-treated samples compared to untreated crude oil. Notably, θ exhibits a positive correlation with CO₂ concentration, clearly indicating that heavy oil viscosity is jointly influenced by both CO₂ and thermal effects, where CO₂ acts synergistically with temperature to enhance viscosity reduction and improve fluidity. A significant observation is that the differential in θ values between CO₂-treated and untreated crude gradually diminishes with increasing temperature. This trend suggests that at elevated temperatures, the spatial structure of heavy oil becomes sufficiently disrupted, thereby reducing CO₂'s relative impact on viscosity. For instance, when dissolved with 40 m³/m³ of CO₂, the heavy oil shows a substantial 53.54% viscosity reduction at 25°C compared to untreated crude, whereas at 80°C, the additional reduction attributable to CO₂ diminishes to merely 0.42%. These findings quantitatively demonstrate that CO₂'s viscosity-reducing effect is most pronounced at lower temperatures, becoming negligible when thermal effects dominate at higher temperatures.

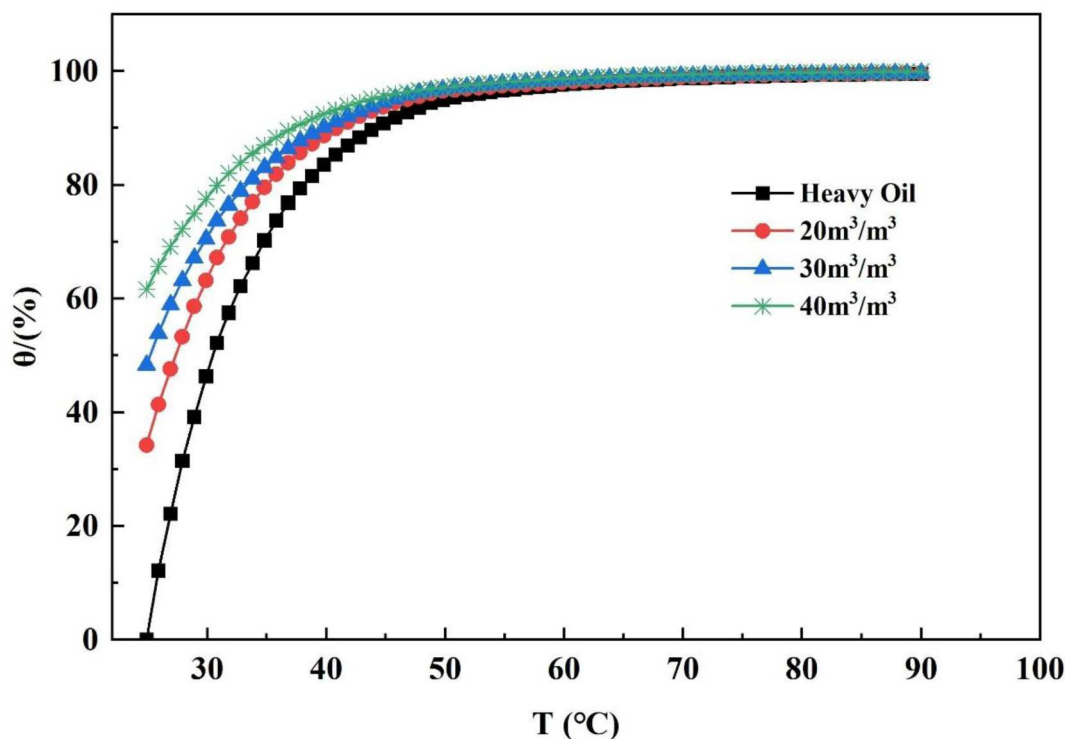


Figure 3—The variation of viscosity reduction ratio with temperature under different CO₂ solubility

To better characterize the synergistic effect between CO₂ and thermal energy, we introduce the viscosity reduction gradient (φ), defined as the change in viscosity reduction ratio (θ) per unit temperature increase at a fixed CO₂ dissolution amount, as expressed in Eq. (2):

$$\varphi = \frac{u_0 - u_1}{\mu_0 \cdot \Delta T} \quad (2)$$

Where, φ is viscosity reduction gradient, %·°C⁻¹; u_0 baseline viscosity at reference temperature T_0 , mPa·s; u_1 is baseline viscosity at reference temperature T_1 , mPa·s; ΔT is absolute temperature change, °C.

Fig. 4 illustrates the viscosity reduction ratio of heavy oil under different CO₂ solubilities. As temperature increases, the viscosity reduction ratio gradually decreases, and the contribution of both temperature and CO₂ to viscosity reduction diminishes. This is primarily because the main mechanism by which temperature and CO₂ reduce heavy oil viscosity involves dissolving resins and asphaltenes, thereby altering the spatial structure of their aggregates (Chen et al. 2022). When the spatial structure of the aggregates is relatively complex, both factors can effectively disrupt this structure, leading to a significant reduction in viscosity. However, once the temperature exceeds a certain threshold, the spatial structure of resin and asphaltene aggregates has already undergone substantial changes, and further temperature increases have a limited effect on modifying the aggregate structure. At lower temperatures, the contribution of both factors to viscosity reduction is more pronounced. Specifically, at 25°C, the viscosity reduction ratio of the crude oil exceeds 12%, whereas at 80°C, it drops to only 0.04%.

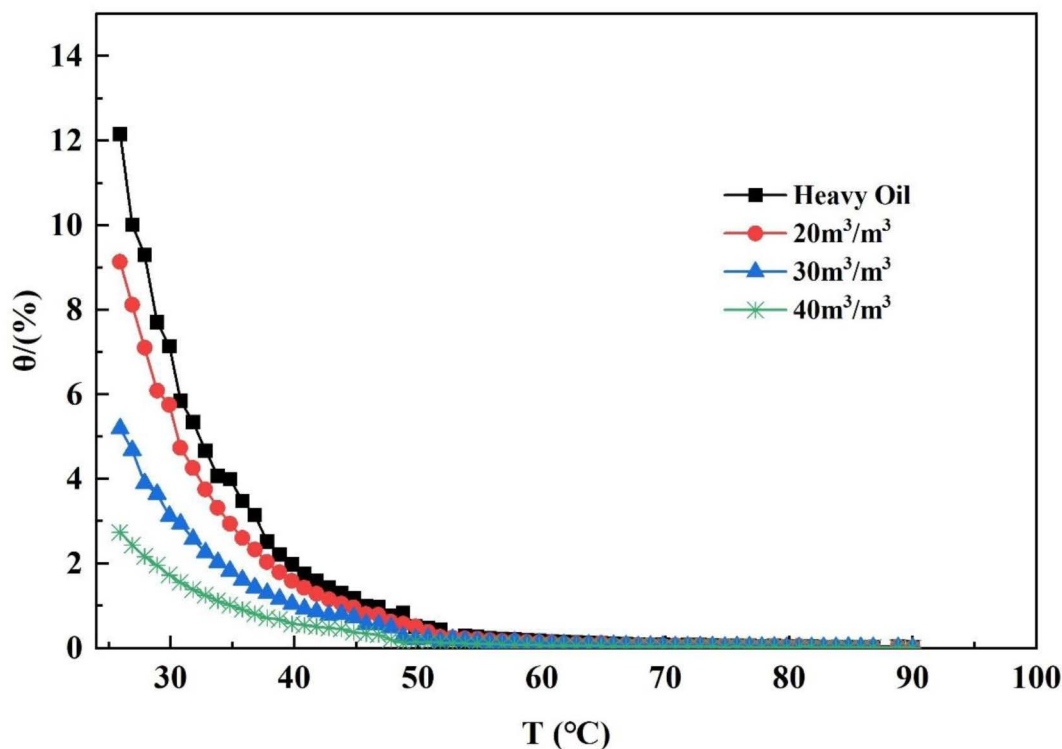


Figure 4—The viscosity reduction rate of crude oil at different mass concentrations.

The Impact of CO₂ on the Rheological Properties of Heavy Oil

As a typical shear-thinning non-Newtonian fluid, heavy oil exhibits an exponential decay in apparent viscosity (η) with increasing shear rate ($\dot{\gamma}$). This phenomenon stems from the dynamic destruction mechanism of shear stress on the three-dimensional network structure of resin-asphaltene aggregates (Hu et al. 2017). As shown in Fig. 5, under isothermal conditions at 25°C, when $\dot{\gamma}$ is relatively low, the supramolecular network formed by hydrogen bonding and π - π stacking within heavy oil demonstrates partial thixotropy, where low shear stress only induces reversible structural deformation. As $\dot{\gamma}$ gradually increases, the shear energy becomes sufficient to overcome intermolecular binding energy, leading to irreversible disaggregation of resin associates and a consequent decrease in apparent viscosity. With increasing CO₂ solubility, the viscosity-shear rate curve of heavy oil shifts significantly downward, indicating that CO₂ synergistically enhances the shear-thinning effect. Notably, under high CO₂ dissolution levels, the $\eta(\dot{\gamma})$ curve tends to flatten, suggesting that solvent molecules reduce the asphaltene volume fraction through a dilution effect while competitively inhibiting the reformation of hydrogen bonds between resin molecules. This process decreases the polarity of the continuous phase, weakening the driving force for structural recovery and thereby diminishing the non-Newtonian characteristics of heavy oil.

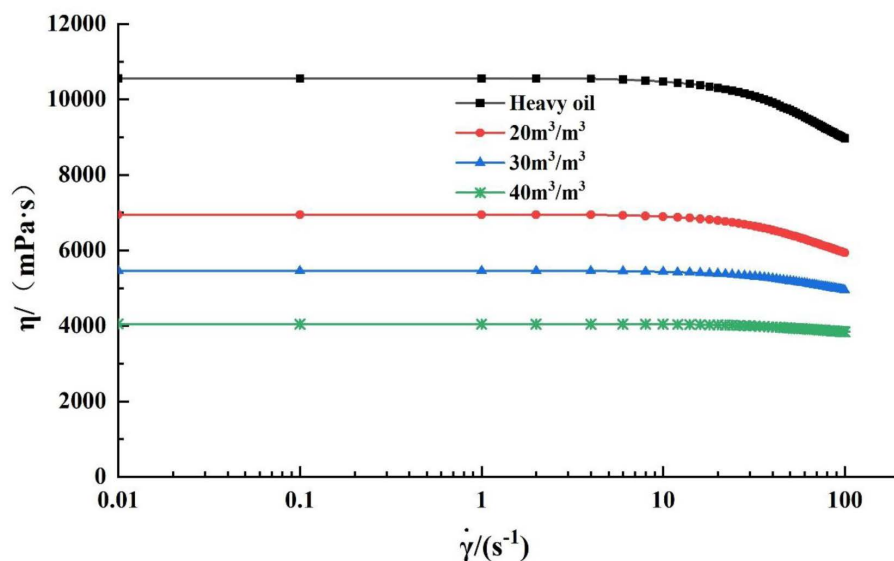


Figure 5—The Relationship between Viscosity and Shear Rate

The rheological phase transition characteristics of CO₂-heavy oil systems can be quantitatively analyzed based on the shear stress (τ)-shear rate ($\dot{\gamma}$) rheological curves (Fig. 6). Experimental results demonstrate that below the anomalous temperature point, heavy oil exhibits typical Bingham plastic fluid behavior, with its rheological curve fitting the Eq. $\tau = \tau_0 + K\dot{\gamma}$, where τ_0 represents the yield stress and K denotes the consistency coefficient. This phenomenon confirms that heavy oil flow must first overcome the elastic energy storage threshold (τ_0) of the resin-asphaltene network structure before triggering continuous shear deformation dominated by viscosity. CO₂ dissolution significantly alters the system's rheological response. As CO₂ solubility increases, the rheological curves systematically shift toward the $\dot{\gamma}$ -axis, accompanied by a decrease in the slope of the fitted line. This indicates that CO₂ weakens the non-Newtonian characteristics of heavy oil, confirming that CO₂ mixing can progressively drive the fluid behavior from shear-thinning toward Newtonian flow.

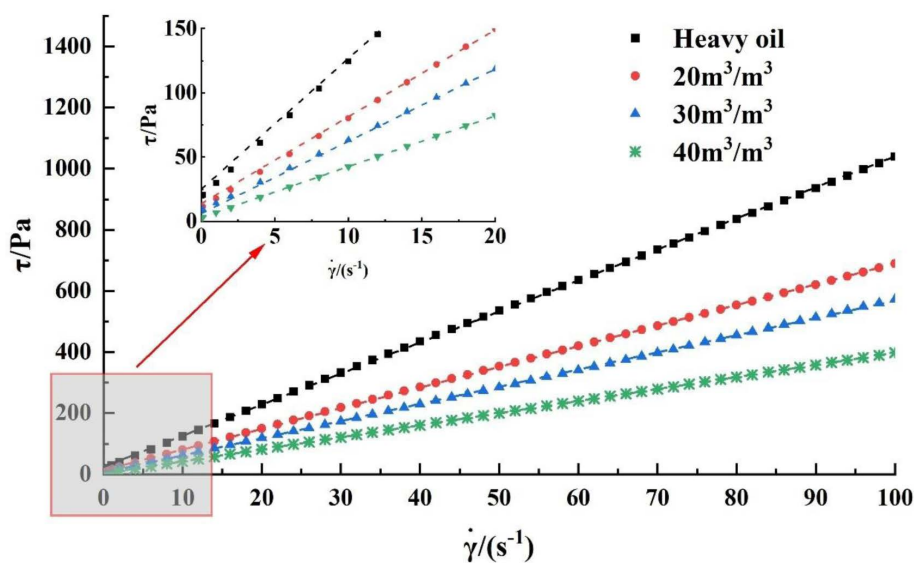


Figure 6—Shear Stress and Shear Rate Relationship

Nonlinear regression analysis of the shear stress-shear rate curves based on the aforementioned rheological model (Table 3) reveals that heavy oil flow behavior requires overcoming an apparent yield stress. Experimental results demonstrate an inverse correlation between the yield stress (τ_0) of heavy oil and CO_2 solubility. When CO_2 solubility increases from 0 m^3/m^3 to 20, 30, and 40 m^3/m^3 , the yield stress (τ_0) of heavy oil decreases from 25.144 Pa to 13.813 Pa, 6.1157 Pa, and 3.1482 Pa, respectively, with a maximum reduction of 87.5%. These results indicate that CO_2 addition effectively reduces the flow resistance of heavy oil, thereby significantly improving its fluidity.

Table 3—Rheological curve constitutive Eq. and yield stress

Crude Oil	Constitutive Eq.	Correlation index	τ_0/Pa
Heavy oil	0	$\tau = 10.155x + 25.144$	99.97
	20	$\tau = 6.5774x + 13.813$	99.99
	30	$\tau = 5.6156x + 6.1157$	99.98
	40	$\tau = 3.9446x + 3.1482$	99.99

Startup Pressure Chart

The variation in crude oil viscosity inevitably affects the PTPG required for flow initiation. When flowing through porous media, heavy oil exhibits strong interactions with pore walls (Wang et al. 2006). The interfacial forces between heavy oil molecules and the viscous forces between the oil and pore surfaces cause the flow to deviate from Darcy's law, manifesting as non-Darcy flow behavior. Flow initiation occurs only when the driving pressure exceeds this threshold pressure (Guo, Li, and Yu 2016)

When CO_2 dissolves into heavy oil, its molecules interact with macromolecular components (e.g., resins and asphaltenes) in the oil, disrupting their π - π stacking and hydrogen-bonding networks, thereby loosening the crude oil structure. As a non-polar gas, dissolved CO_2 dilutes the polar components (such as asphaltenes and resins) in the crude oil, reducing the viscoelastic modulus of the system. The viscosity-reducing effect intensifies with increasing CO_2 concentration. According to Darcy's law, fluid flow resistance is positively correlated with viscosity; thus, viscosity reduction significantly enhances percolation efficiency. Meanwhile, the decrease in the pseudo-threshold pressure gradient means that, under the same pressure conditions, crude oil can more easily overcome capillary resistance in pore throats, thereby improving displacement efficiency. Furthermore, CO_2 dissolution enhances heavy oil percolation through the following mechanisms. Volumetric Expansion Effect: CO_2 dissolution induces oil swelling, lowering the concentration of polar substances per unit volume and weakening oil-rock adsorption forces. Structural Disruption: CO_2 breaks down the spatial network structure of heavy oil macromolecules, reducing the size of resin-asphaltene aggregates, which further improves percolation capability.

Fig. 7 reveals the influence of CO_2 dissolution volume on the pseudo-threshold pressure gradient (PTPG) of heavy oil. Experimental data demonstrate that the threshold pressure data points under different shear rates conform to the Bingham fluid model. The intercept of the fitted curve on the positive half-axis of the threshold pressure axis represents the PTPG value (G) under specific temperature conditions. As temperature increased from 40°C to 80°C, the PTPG gradually decreased. For instance, the G value of heavy oil without dissolved CO_2 dropped from 4.63 MPa/m to 0.09 MPa/m, indicating that elevated temperature significantly improves fluidity by reducing the phase-state viscosity of heavy oil. Under a constant temperature (40°C), when CO_2 solubility rose from 0 m^3/m^3 to 40 m^3/m^3 , the G value decreased from 4.63

MPa/m to 0.14 MPa/m. This implies that at 40°C, a CO₂ solubility of 40 m³/m³ achieves flow performance comparable to heating to 80°C, substantially reducing thermal energy requirements. Mechanistic analysis indicates that the synergistic effect of temperature and CO₂ alters the rheological properties of heavy oil. Temperature elevation reduces internal cohesion by enhancing molecular thermal motion. CO₂ dissolution weakens structural strength through dilution effects and interfacial tension modulation. This dual-action mechanism provides a theoretical foundation for designing thermal-chemical hybrid flooding strategies in heavy oil reservoirs.

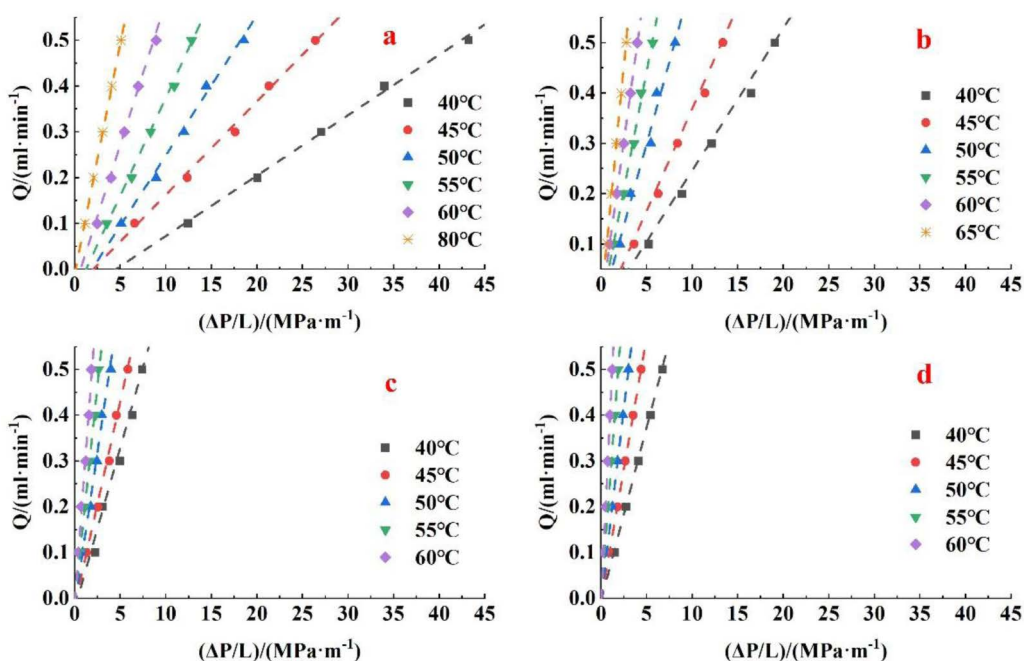


Figure 7—Threshold pressure gradient curve of heavy oil under different solubility

Based on the fitted curves in Fig. 7, we systematically integrated the threshold pressure gradient data of heavy oil under different temperatures and CO₂ dissolution conditions, and established a log-log relationship between the pseudo-threshold pressure gradient (G) and oil mobility (λ), as shown in Fig. 8. The experimental results demonstrate that this relationship follows a power-law function ($G = a\lambda^b$, see Eq. 3), where parameters a and b represent the baseline flow resistance and mobility sensitivity, respectively. When oil mobility falls below the fitted curve, the heavy oil becomes immobile. To quantitatively evaluate the mobilization effect of CO₂, we calculated the area under the fitted curve by integrating Eq. 2, with the lower limit set as the minimum mobility corresponding to different solubilities at reservoir temperature and the upper limit as $+\infty$. The results show that the proportion of immobile oil decreased significantly from 47.09% (without CO₂) to 18.31%, 6.37%, and 1.79% with CO₂ addition, representing reduction rates of 61.13%, 86.47%, and 96.20%, respectively. These findings further confirm the effectiveness of CO₂ in enhancing heavy oil producibility through viscosity reduction and mobility improvement mechanisms. The study provides important insights for optimizing thermal-chemical hybrid flooding strategies in heavy oil reservoirs.

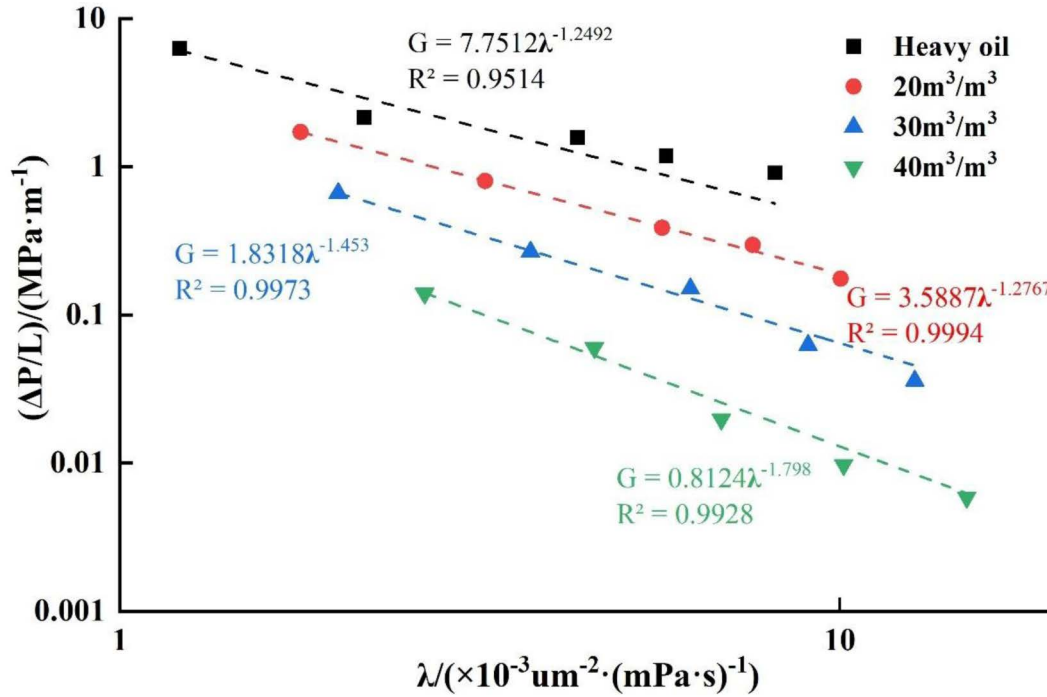


Figure 8—Flowability limit diagram of heavy oil after adding different solvents

$$G = \begin{cases} 7.7512\left(\frac{k}{\mu}\right)^{-1.2492}, R^2 = 0.9514, \text{ Heavy Oil;} \\ 3.5887\left(\frac{k}{\mu}\right)^{-1.2767}, R^2 = 0.9994, 20\text{m}^3 / \text{m}^3; \\ 1.8318\left(\frac{k}{\mu}\right)^{-1.453}, R^2 = 0.9973, 30\text{m}^3 / \text{m}^3; \\ 0.8124\left(\frac{k}{\mu}\right)^{-1.798}, R^2 = 0.9928, 40\text{m}^3 / \text{m}^3. \end{cases} \quad (3)$$

By substituting Eq. (3) into the formula for ultimate drainage radius (Eq. 4), we obtain the relationship (Eq. 5) between the ultimate drainage radius (R_e) of this block and oil mobility as well as production pressure differential. This Eq. quantitatively reveals how CO_2 enhances the ultimate drainage capacity of heavy oil reservoirs.

$$R_e = \frac{P_e - P_w}{\Delta P / L} \quad (4)$$

$$R_e = \begin{cases} 0.129 \cdot (P_e - P_w) \cdot \lambda^{1.2492}, \text{ Heavy Oil;} \\ 0.2787 \cdot (P_e - P_w) \cdot \lambda^{1.2767}, 20\text{m}^3 / \text{m}^3; \\ 0.5459 \cdot (P_e - P_w) \cdot \lambda^{1.453}, 30\text{m}^3 / \text{m}^3; \\ 1.2309 \cdot (P_e - P_w) \cdot \lambda^{1.798}, 40\text{m}^3 / \text{m}^3. \end{cases} \quad (5)$$

In thermal recovery of heavy oil reservoirs, the scientifically determined temperature threshold for drive mechanism transition must ensure that the in-situ crude oil mobility reaches a critical value to guarantee effective flow capacity. Otherwise, the poor flow capacity of heavy oil would lead to unsatisfactory development performance. Therefore, the activation temperature of heavy oil constitutes a crucial parameter in development strategies. By fitting the logarithm of activation pressure against temperature, we established the activation temperature chart shown in Fig. 9, with the fitting Eq. given by Formula (6). The results demonstrate that as CO_2 solubility increases, the required activation temperature to achieve the same activation pressure progressively decreases. Based on this relationship, oilfields can first determine the target flow capacity required for in-situ crude oil according to actual production needs, then use this formula to

calculate the precise temperature requirement, thereby optimizing the thermal input needed for production and development.

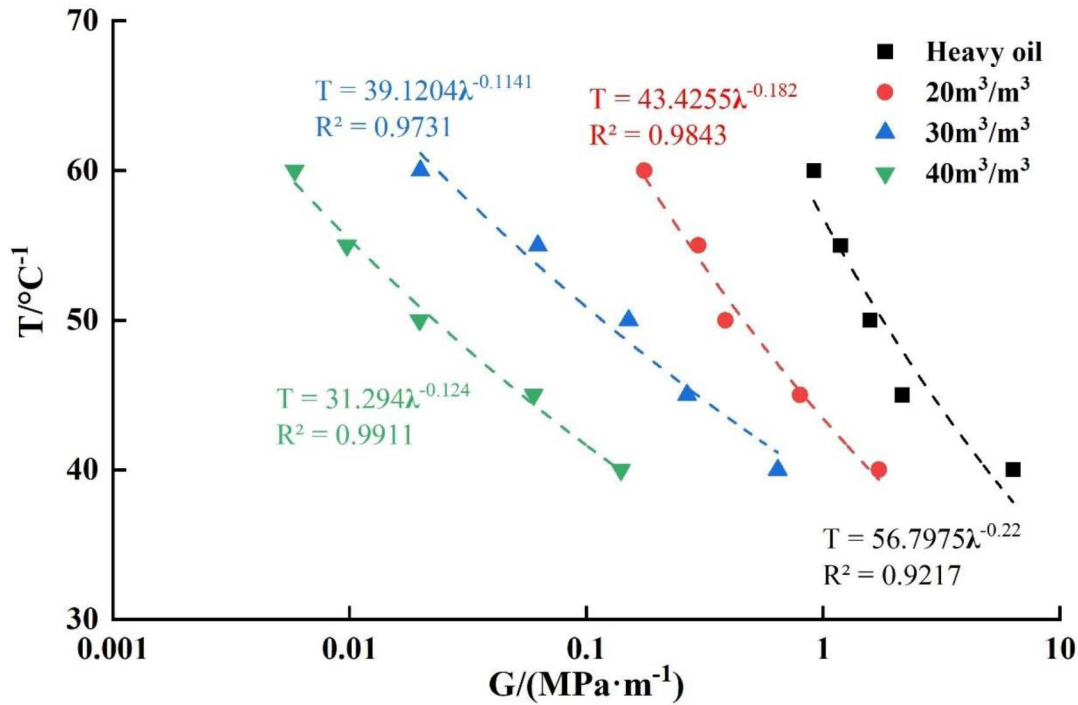


Figure 9—Threshold temperature limit diagram of heavy oil after adding different solvents

$$T = \begin{cases} 56.7975\left(\frac{k}{\mu}\right)^{-0.22}, R^2 = 0.9217, \text{ Heavy Oil;} \\ 43.4255\left(\frac{k}{\mu}\right)^{-0.182}, R^2 = 0.9843, 20\text{m}^3 / \text{m}^3; \\ 39.1204\left(\frac{k}{\mu}\right)^{-0.1141}, R^2 = 0.9731, 30\text{m}^3 / \text{m}^3; \\ 31.294\left(\frac{k}{\mu}\right)^{-0.124}, R^2 = 0.9911, 40\text{m}^3 / \text{m}^3. \end{cases} \quad (5)$$

Conclusions

This study focuses on heavy oil from a block in Xinjiang Oilfield as the research subject. By integrating the non-Newtonian fluid characteristics and nonlinear flow behavior of heavy oil, we systematically investigate the intrinsic mechanisms through which CO₂ enhances the flow capacity of heavy oil in porous media. The main conclusions are as follows:

1. The introduction of CO₂ significantly reduces heavy oil viscosity, with the viscosity reduction effect intensifying as CO₂ dissolution increases. However, the viscosity reduction ratio gradually decreases with rising temperature, indicating that the contributions of both temperature and CO₂ to viscosity reduction diminish progressively.
2. CO₂ injection effectively reduces both the anomalous temperature and yield stress of heavy oil. Experimental measurements show that when CO₂ dissolution reaches 20, 30, and 40 m³/m³, the anomalous temperature decreases to 49.8°C, 47.8°C, and 45.7°C respectively.
3. CO₂ dissolution significantly enhances the producibility of formation heavy oil. With CO₂ dissolution amounts of 20, 30, and 40 m³/m³, the immobile oil volume fraction decreases by 61.13%, 86.47%, and 96.20%, respectively. Based on this model, we established ultimate drainage radius Eq.s for heavy

oil under varying CO₂ dissolution conditions, providing a theoretical foundation for well spacing optimization design.

4. This study developed a prediction model for heavy oil activation temperature based on rheological-thermodynamic coupling, enabling quantitative design of development temperature parameters. The model establishes an explicit relationship between crude oil activation pressure and activation temperature (T), revealing the temperature requirement patterns under various development conditions.

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